

2D SCANNING LIDAR · LR-1BS SERIES

30-METRE OLEI LR-1BS5+ 270° LIDAR

The longest reach in the LR-1BS family. Target enhancement technology pushes a 10% reflectivity target out to 15 metres and an 80% target to 30 — in the same 450-gram, 60 × 60 × 85 mm body as the rest of the series.



■ WHY THIS SENSOR

- **0.2–30 m at 80% reflectivity** — the longest range in the LR-1BS family, and 30 m on reflectors too
- **Target enhancement technology** — a 10% reflectivity target reaches 15 m — nearly double the standard LR-1BS5
- **Built for asphalt and plant rooms** — the low-reflectivity surfaces that defeat a standard scanner
- **User-settable weak-signal threshold** — tune sensitivity to the site without changing the mount
- **Built-in multi-level filtering** — configurable tailing suppression at the sensor, so the cloud arrives clean
- **< ±20 mm absolute accuracy on common targets** — < ±40 mm on reflectors, ≤ 20 mm repeat
- **40K points per second, 0.225° fixed resolution** — azimuth, distance and reflection intensity over 100 Mbps Ethernet
- **450 g, IP65, 5 W at 12–24 V DC** — M8 power and LAN on 2 m leads, OLAMViewer and a ROS driver

0.2–30 m

MEASUREMENT RANGE

15 m

AT 10% REFLECTIVITY

< ±20 mm

ABSOLUTE ACCURACY

40K

POINTS / SECOND

■ APPLICATION SCENARIOS



SLAM NAVIGATION

Point-cloud mapping and localisation for mobile platforms



AMR & AGV

270° obstacle avoidance from a single chassis-mounted scanner



INTRALOGISTICS

Forklifts, tuggers and conveyor zones in active aisles



DOCKING & TRACKING

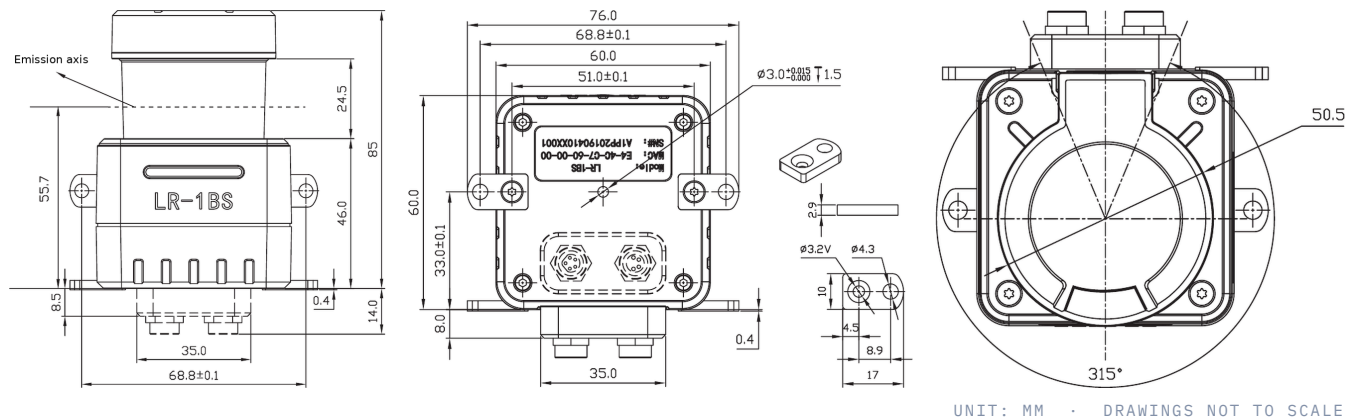
Reflector-based docking and pallet or object tracking

LR-1BS5+ — SPECIFICATIONS

PARAMETER	TECHNICAL DATA
Sensing technology	2D scanning ToF LiDAR
Laser wavelength	905 ± 20 nm
Laser safety level	Class 1 (IEC 60825-1:2014)
Laser channels	1
Field of view (horizontal)	270°
Beam divergence	5.8 mrad longitudinal 0.58 mrad transversal
Scanning frequency	10–25 Hz
Angle resolution (horizontal)	0.225°
Measurement range (based on reflectivity)	0.2–30 m @ 80% reflectivity 0.2–15 m @ 10% · 0.2–5 m @ 4% · 0.2–30 m @ reflector
Accuracy ①	Common targets: absolute < ±20 mm Reflector: absolute < ±40 mm · Repeat ≤ 20 mm
Distance resolution	1 mm
Signal strength	0–60000
Scanning rate	16K–40K pts/s
Horizontal error (horizontal angle)	≤ ±1°

PARAMETER	TECHNICAL DATA
Switching outputs	1
Ethernet connection	100 Mbps Ethernet
Connection	5-pin M8×1 female, 2 m (power) 4-pin M8×1 female, 2 m LAN
Operating voltage	12–24 V DC
Power consumption @ 25 °C	5 W
Mechanical dimensions	60 × 60 × 85 mm
Weight	450 g
Degree of protection	IP65 or optional
Operating temperature	–10 to +50 °C
Storage temperature	–30 to +70 °C
Air humidity	< 95% RH
Optical indicators	RGB × 4
Ambient light limit	80 000 lux
Mounting	Bottom / back
Basic software	OLAMViewer · ROS driver

OUTLINE AND MOUNTING



Certification. FCC, EMC, CE, RoHS, REACH, FDA. EMC per IEC 61000-6-2:2016-08 and IEC 61000-6-3:2006-07. Vibration per IEC 60068-2-6:2007. Shock per IEC 60068-2-27:2008.

Functional safety. The manufacturer's specification for this product states laser safety Class 1 per IEC 60825-1:2014 and the EMC, vibration and shock standards listed above. It does not state an IEC 61496 Type rating or an ISO 13849 performance level. Treat the zone outputs as obstacle-avoidance and area-monitoring signals, not as a rated safety function, and do not use this sensor as the sole protective device. Where a rated scanner is required, ask MorpheusTEK about the OLEI GS1-5.

Notes. ① Standard value; actual value depends on environmental conditions. The 15 m figure at 10% reflectivity is the reason to specify this variant over the LR-1BS5 — confirm it against the actual surfaces on site, because target enhancement behaviour depends on surface roughness as well as reflectivity. Weak-signal threshold and filter level are user settings; record the values used when a map is commissioned.

Installation. Mount the scanner so the 270° plane is unobstructed through its full sweep and the blind 90° faces structure, not traffic. Keep the optical window clear of spray and dust and clean it on a schedule in dirty environments. Re-run the map after any change to mounting height or angle.

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